Beyond Factual QA: Extending the Low-Resource Hallucination-Detector Robustness Phenomenon to Dialogue Generation and a Broader Set of Languages

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Problem statement

Datta et al. (EACL 2026) found that when large language models move from English to low-resource languages, task accuracy on short factual questions falls sharply while three hallucination detectors (MAM, SelfCheckGPT, and Semantic Entropy) keep most of their ranking ability. They measure this with TPHR, the log-ratio of the accuracy drop to the detector's AUROC drop relative to English. Their evidence covers five languages and one response format: a single short factual answer per question.

The paper names its own two biggest limitations directly: the language set is narrow, and dialogue is left as future work because a single reply can contain several hallucinations rather than one. Both point at the same open question: does the detector keep working because hallucination detection is genuinely easier than generation, or because the setting (five languages, one-line answers) happened to be forgiving? We test this along both axes: a much wider and more typologically varied set of languages, and answers long and unconstrained enough that a hallucination has to be located inside them rather than judged as a single yes/no. If the accuracy-versus-AUROC gap narrows or disappears under either extension, the base paper's finding is a property of short factual QA rather than of hallucination detection generally, and that matters for real deployments, where models write full answers in dozens of languages, most with little training data.

Dataset

Three datasets, each doing one job:

- **mTREx**: the base paper's own data and released detector code (English, German, Hindi, Bengali, Urdu; short factual questions on capitals, countries, official languages). Used as the reference point our results are measured against.

- **X-FACTR**: a cloze-style factual probing benchmark covering 23 languages, from well-resourced ones down to genuinely low-resource languages such as Cebuano, Yoruba, Ilokano and Swahili (Jiang et al., EMNLP 2020). Supplies the breadth axis: languages further from English than Bengali or Urdu.
- **PsiloQA**: a grounded, multilingual QA dataset covering 14 languages, where each item is a full model-generated answer with the hallucinated span (if any) marked at the character level, built from 26 different model checkpoints (Rykov et al., Findings of EMNLP 2025). Supplies the depth axis: answers long and unconstrained enough that a detector has to catch an error inside free-form text, not just judge one short fact.

PsiloQA is single-turn grounded QA, not multi-turn dialogue. No multilingual dataset with comparable hallucination labels exists at multi-turn scale. It is the closest available stand-in for the base paper's dialogue axis and is treated as such throughout; genuine multi-turn dialogue is flagged as out of scope below rather than implied by the project title.

Paper link

Datta, Chilukuri, Kumar, Ghosh & Zafar (2026). *Do LLM hallucination detectors suffer from low-resource effect?* EACL 2026. https://aclanthology.org/2026.eacl-long.136/

Dataset sources: mTREx: https://github.com/aisoc-lab/low-resource-hallucination-detection · X-FACTR: https://x-factr.github.io/ · PsiloQA: https://huggingface.co/datasets/s-nlp/PsiloQA

Scope of the project

In scope

- Reusing the base paper's three detectors unchanged (MAM, SelfCheckGPT, Semantic Entropy) and its TPHR metric, so results are directly comparable to the numbers already published.
- Breadth test: running the same detector pipeline on a stratified subset of X-FACTR languages spanning high-, medium- and low-resource tiers, scored the same way as mTREx.
- Depth test: running the same detectors on PsiloQA, binarizing each item as hallucinated or not from its span annotation, and computing AUROC and TPHR the same way as the base paper.
- One open-weight model family already used in the base paper (final choice fixed in Phase 2 once hardware is confirmed), held constant across both tests so any difference in the accuracy-versus-AUROC gap comes from the data, not the model.

- A comparative analysis of TPHR across three settings: the base paper's own five languages, the X-FACTR breadth test, and the PsiloQA depth test.

Out of scope

- Designing a new detector: the project tests whether the base paper's three existing detectors hold up, not whether a better one exists.
- Multi-turn dialogue data collection or annotation.
- Calibration analysis (ECE) or any metric beyond accuracy, AUROC and TPHR.
- Running all 23 X-FACTR languages or the full model set from the base paper; the exact language subset and model choice are finalized in Phase 2 against available compute.

Deliverable (Track B): a short comparative-analysis report and a codebase, built on the base paper's own released code, that reports whether its accuracy-versus-detector-robustness gap holds, narrows, or closes under a broader language set and longer, less constrained generation.